FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2016**

HALF COURSE TEST – VI

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 04)** contains 4 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (05 – 08) contains 4 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

   Section-A (09 to 11) contains 1 paragraph having 3 questions. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.

3. **Section-C (01 – 06)** contains 6 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

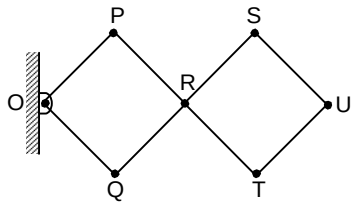
Single Correct Choice Type

*This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

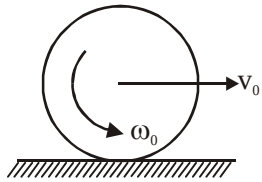
1. Block A and B have masses 40 kg. and 60 kg resp. They are placed on a smooth surface and the spring connected between them is stretched by 1.5m. If they are released from rest, determine the speeds of the blocks at the instant the spring became unstrentched
   (A) $V_1 = 3$ m/s, $V_2 = 4.5$ m/s (B) $V_1 = 4.5$ m/s, $V_2 = 3$ m/s
   (C) $V_1 = 5$ m/s, $V_2 = 6$ m/s (D) $V_1 = 4$ m/s, $V_2 = 3$ m/s

2. The end of a chain of length L and mass per unit length ρ, which is piled up on a horizontal platform is lifted vertically with a constant velocity u by a variable force F. Find F as a function of height x of the end above platform.
   (A) $\rho\left(gx + 2u^2\right)$ (B) $\rho\left(2gx + \rho u^2\right)$
   (C) $\rho\left(gx + u^2\right)$ (D) $\rho\left(u^2 - gx\right)$

3. In the arrangement of rigid links of equal length l, they can freely rotate about the joined ends as shown in the figure. If the end U is pulled horizontally with constant speed 20 m/s, find the approx. speed of end P when the angle SUT is 90°.
   (A) 5 m/s (B) 10 m/s
   (C) 7.1 m/s (D) 14.12 m/s



4. A disc of radius R is spun to an angular speed ω_0 about its axis and then imparted a horizontal velocity of magnitude $\frac{\omega_0 R}{4}$ (at t = 0) with its plane remaining vertical. The coefficient of friction between the disc and the plane is μ. The sense of rotation and direction of its linear speed are shown in the figure. Choose the correct statement. The disc will return to its initial position
   (A) if the value of $\mu < 0.5$. (B) irrespective of the value of μ ($\mu > 0$).
   (C) if the value of $0.5 < \mu < 1$. (D) if $\mu > 1$.



Space for rough work

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

5. The figures represent two snaps of a travelling wave on a string of mass per unit length, μ= 0.25kg/m. The two snaps are taken at time t = 0 and at $t = \frac{1}{24}$ s. Then

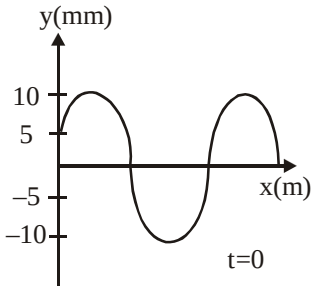


Figure -1

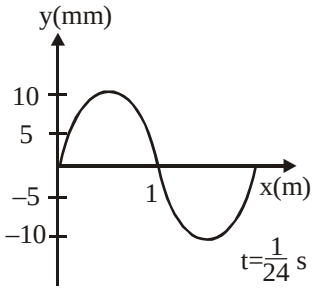


Figure -2

(A) speed of wave is 4 m/s

(B) the tension in the string is 4 N

(C) the equation of the wave is $y = 10 \sin (\pi x - 4\pi t + \frac{\pi}{6})$

(D) the maximum velocity of the particle $= \frac{\pi}{25} m/s$

6. An ideal gas undergoes a process such that $P \propto \frac{1}{T}$. The molar heat capacity of this process is 33.24 J/mol K.

(A) The work done by the gas is $2R\Delta T$.

(B) Degree of freedom of the gas is 4.

(C) Degree of freedom of the gas is 3.

(D) $Y \left(\frac{C_P}{C_V}\right)$ for the gas is 1.5.

Space for rough work

7. Velocity of a particle of mass 2 kg changes from $\vec{v}_1 = -2\hat{i} - 2\hat{j}\ m/s$ to $\vec{v}_2 = (\hat{i} - \hat{j})m/s$ after colliding with a plane surface

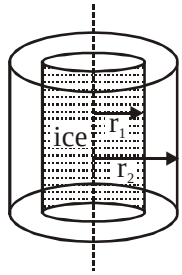
(A) the angle made by the plane surface with the positive x–axis is $90^o + \tan^{-1}\left(\frac{1}{3}\right)$

(B) the angle made by the plane surface with the positive x–axis is $\tan^{-1}\left(\frac{1}{3}\right)$

(C) the direction of change in momentum makes an angle $\tan^{-1}\left(\frac{1}{3}\right)$ with the +ve x–axis

(D) the direction of change in momentum makes an angle $90^\circ + \tan^{-1}\left(\frac{1}{3}\right)$ with the plane surface

8. A 100 cm long cylindrical flask with inner and outer diameters 2 cm and 4 cm respectively is completely filled with ice as shown in the figure. The constant temperature outside the flask is 40ºC. (Thermal conductivity of the flask is 0.693 W/mºC, L_{ice} = 80 cal/gm).



(A) Rate of heat flow from outside to the flask is 80π J/s.

(B) The rate at which ice melts is $\frac{\pi}{4200}$Kg / s .

(C) The rate at which ice melts is 100π Kg/s.

(D) Rate of heat flow from outside to flask is 40 π J/s.

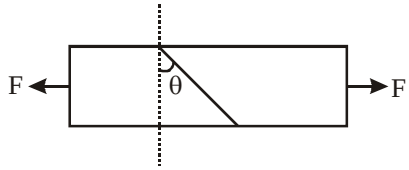
Space for rough work

Comprehension Type

*This section contains **one group** of questions. This group has **3 multiple** choice question based on a paragraph. This question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct*

Paragraph for Question Nos. 9 to 11

A bar of cross-section A is subjected to equal and opposite tensile forces F at its ends. Consider a plane through the bar making an angle θ with a plane at right angles to the bar as shown in figure



9. The tensile stress at this plane in terms of F, A and θ is

(A) $\frac{F\cos^2\theta}{A}$ (B) $\frac{F}{A\cos^2\theta}$

(C) $\frac{F\sin^2\theta}{A}$ (D) $\frac{F}{A\sin^2\theta}$

10. In the above problem, for what value of θ is the tensile stress a maximum?

(A) Zero (B) 90°

(C) 45° (D) 30°

11. The shearing stress at the plane, in terms of F, A and θ is

(A) $\frac{F\cos 2\theta}{2A}$ (B) $\frac{F\sin 2\theta}{2A}$

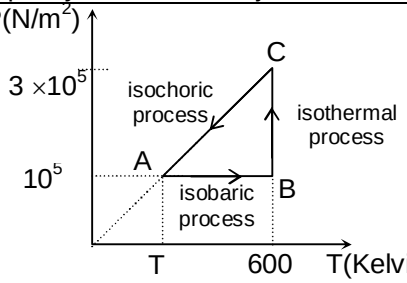
(C) $\frac{F\sin\theta}{A}$ (D) $\frac{F\cos\theta}{A}$

Space for rough work

SECTION – B
(Matching List Type)

*This section contains **2 questions.** Each question has four statements (A, B, C and D) given in **Column I** and **five statements** (p, q, r, s and t) in **Column II.** Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.*

1. Column-I contains some thermodynamic system and column-II contains its properties. Match the following columns.

	Column – I		Column – II
(A)	A steel rod is clamped at its two ends and rests on a fixed horizontal base. The longitudinal strain developed in the rod when temperature rises to 50^0C is -3.6×10^{-4}, and the temperature coefficient of linear expansion of steel is $1.2\times10^{-5}/^0$C. Find the temperature at which rod is in relaxed state	(p)	400
(B)	Six hundred mole of a gas undergoes a process $P=\dfrac{7\times10^5 N/m^2}{2+\dfrac{V}{4m^3}}$ where V is volume in m^3. When volume of the gas is 20 m^3, its temperature in Kelvin is approximately equal to	(q)	27
(C)	A block of mass 100 gm slides on a rough horizontal surface of zero heat capacity. When the speed of the block decreases from 10 m/s to 5 m/s, its temperature rises by 0.25^0C. The heat capacity of the block in joule/°C is	(r)	20
(D)	 If two moles of a gas consisting of 1 mole of He and 1 mole of H_2 undergo a cyclic process $A\rightarrow B\rightarrow C\rightarrow A$ as shown in the figure the total work done in the cycle is $-600R\ln[9/e]$ The temperature T in ^{0}C is	(s)	15
		(t)	None of these

Space for rough work

2. A standing wave, formed in an organ pipe aligned along the x-axis, is given by (displacement)y = (5×10^{-6} m)sin(4πx) cos (1320πt) where x is in metre and t is in seconds.

	Column – I		**Column – II**
(A)	Change in pressure is maximum	(p)	x = 1/4 m
(B)	Change in pressure is minimum	(q)	x = 1/2 m
(C)	Maximum amplitude of vibration	(r)	x = 1 m
(D)	Minimum amplitude of vibration	(s)	x = 1/8 m
		(t)	x = 3/8 m

SECTION – C

Integer Answer Type

*This section contains **6** questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.*

1. The equation of a transverse wave is given by

$$\psi = 10^{-2}\sin\pi[30t - x\sqrt{3} - y]$$

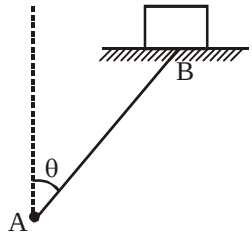
where x, y and ψ are in metre, and t in second.
If phase difference between two points
$A(2\sqrt{3}m, 2m)$ and $B(3\sqrt{3}m, 3m)$ be $n\pi$. Find value of n

2. The emissivity of tungsten is approximately 0.35. A tungsten sphere 1 cm in radius is suspended within a large evacuated enclosure whose walls are at 300 K. What power input is required to maintain the sphere at a temperature of 3000 K if heat conduction along the supports is neglected? Express your answer in kW after rounding off the nearest integer. Take $\sigma = \frac{17}{3}\times 10^{-8}$ S. I. units and $\pi = \frac{213}{68}$.

Space for rough work

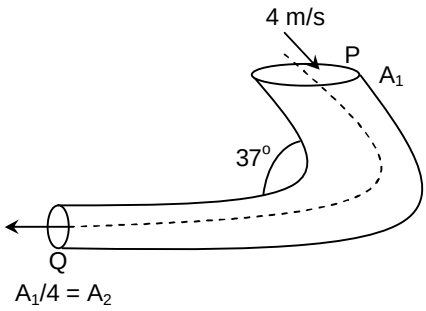
3. Maximum height reached by a rocket fired with a speed equal to 50% of the escape velocity from earth's surface is given by R/x. Calculate value of x ? Where R is radius of earth.

4. A rod AB of length 2 m is hinged at point A and it's other and B is attached to a platform on which a block of mass m is kept. Rod rotates about point A maintaining angle θ = 30º with the vertical in such a way that platform remains horizontal and revolves on the horizontal circular path. If the coefficient of static friction between the block and platform is μ = 0.1 then find the maximum angular velocity in rad/s of rod so that block does not slip on the platform. [g = 10 m/s^2]



5. A cylinder of mass M radius R is resting on a horizontal platform (which is parallel to x–y plane) with its axis fixed along the y–axis and free to rotate about its axis. The platform is given a motion in X–direction given by X = A cos ωt. There is sufficient friction present in the surface of contact that can prevent the slipping between the cylinder and platform. The minimum torque in N-m acting on the cylinder during its motion is.
[Take M = 4 kg , R = 1m, A = 2m, ω = 1 rad/s]

6. Oil enters the bend of a pipe in the horizontal plane with velocity 4 m/s & pressure $280\times10^3 N/m^2$ A1/4=A_2 as shown. If pressure at Q in (KN/m^2) is $N\times 43$. Find N (specific gravity of oil 0.9 & $\cos 37 = 0.8$)



Space for rough work

Chemistry

PART - II

SECTION – A

Single Correct Choice Type

*This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. An inorganic compound (X) on hydrolysis produces a gas which on treatment with sodium followed by its reaction with ethyl chloride gives another compound (Y). Compound (Y) on heating with P – 2 catalyst gives (2Z) – pent – 2 – ene as major product. Hence the inorganic compound (X) is
   (A) Tl_4C (B) BaC_2
   (C) SiC (D) Mg_2C_3

2. Boron trichloride on heating with ammonium chloride at 140°C gives a compound (A) which on reduction with sodium borohydride followed by hydrolysis gives another compound (B). Hence compound (A) and (B) are respectively.
   (A) inorganic graphite & ammonia (B) inorganic benzene & orthoboric acid
   (C) trichloroborazine & orthoboric acid (D) diborane & orthoboric acid

3. (S) – 2 – chloropentane on reaction with Cl_2 (g) in the presence of U.V. light, gives 'N' number of optically active dichloro derivatives. Hence the value of N is
   (A) 5 (B) 6
   (C) 7 (D) 4

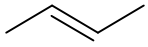
4. 3 mole of KIO_3, 4 mole of KI and 5 mole of HCl are allowed to react according to the equation given below:
   $KIO_3 + 2KI + 6HCl \longrightarrow 3ICl + 3KCl + 3H_2O$
   The number of moles of products are
   (A) 27 (B) 18
   (C) 7.5 (D) 9.5

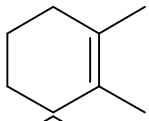
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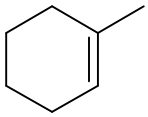
Multiple Correct Answers Type

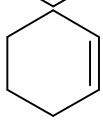
*This section contains 4 multiple correct answer(s) type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE** is/are correct.*

5. Which of the following compounds will give same product in hydroboration oxidation and catalytic hydration?

(A) 

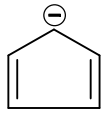
(B) 

(C) 

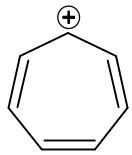
(D) 

6. Which of the following are aromatic in character?

(A) 

(B) 

(C) 

(D) 

7. Which of the following compounds undergo hydrolysis?

(A) CCl_4 (B) $SiCl_4$
(C) BCl_3 (D) SiH_4

8. Which of the following statement (s) is/are correct regarding photoelectric effect?
   (A) No electrons are ejected, regarding of the intensity of the radiation, unless the frequency exceeds a threshold value characteristic of the metal
   (B) The kinetic energy of the ejected electrons increases linearly with the frequency of incident radiation but is independent of the intensity of the radiation.
   (C) Even at low intensities, electrons are ejected immediately if the wavelength is above threshold value.
   (D) It is a direct evidence for the dual nature of light.

Space for rough work

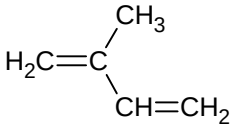
Linked Comprehension Type

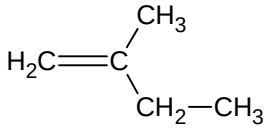
*This section contains one group of questions. This group has 3 multiple choice question based on a paragraph. This question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct*

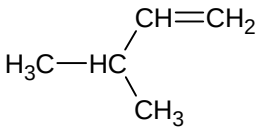
Paragraph for question Nos 9 to 11

A more widely used method for locating the double bond of alkene involves the use of ozone. Ozone reacts with alkene to form unstable compounds called ozonide. Ozonides on reduction with Zn and water produces carbonyl compound that can be isolated and identified. Ozonolysis can be either of reductive type or oxidative type.

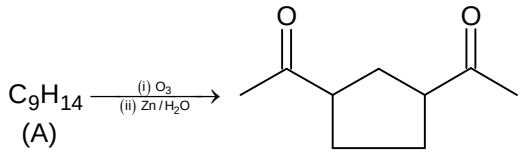
9. An organic compound A (C_5H_8) on hydrogenation gives compound B (C_5H_{12}). A on ozonolysis gives methanal and 2 – ketopropanal. The compound (A) is

(A) 

(B) 

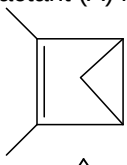
(C) 

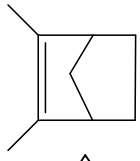
(D) $H_3C-CH=CH-CH_2-CH_3$

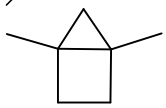
10. 

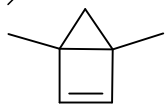
(A)

The reactant (A) is

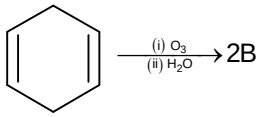
(A) 

(B) 

(C) 

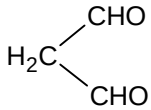
(D) 

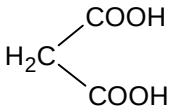
Space for rough work

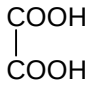
11. 

The product B is

(A) 

(B) 

(C) 

(D) 

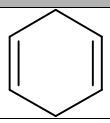
SECTION – B
(Matrix-Match Type)

*This section contains **2 questions.** Each question has four statements (A, B, C and D) given in **Column I** and **five statements** (p, q, r, s and t) in **Column II.** Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.*

1. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	I_3^-	(p)	Linear
(B)	XeO_3	(q)	Square planar
(C)	$Ni(CN)_4$	(r)	Trigonal bipyramidal
(D)	OSF_4	(s)	sp^3
		(t)	dsp^2

2. Match the Column – I with Column – II:

Column – I (Compounds)		Column – II (Ozonolysis products)	
(A)		(p)	Formaldehyde
(B)	$CH_2 = CH – CH = CH – CH_3$	(q)	Glyoxal
(C)	$(CH_3)_2C = CH – CH = CH_2$	(r)	Propane – 1, 3 – dial
(D)	$CH_3 – CH = CH_2$	(s)	Acetone
		(t)	Acetaldehyde

Space for rough work

SECTION – C

(Integer Answer Type)

*This section contains **6** questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.*

1. One litre of saturated solution contains AgSCN and AgBr. Calculate the ratio of $\frac{\left[SCN^-\right]}{\left[Br^-\right]}$ in the solution. $\left[K_{sp} \text{ of } AgSCN = 1.0\times10^{-12}; K_{sp} \text{ of } AgBr = 5\times10^{-13}\right]$

2. In the following reaction we start with 2 mol of N_2 and 5 mol of H_2 exerting a total pressure of 7 atm at a given temperature in a closed vessel when 50% of N_2 is converted into NH_3, then
$N_2 + 3H_2 \longrightarrow 2NH_3$
What will be the partial pressure of NH_3 in atmosphere?

3. If uncertainities in measurement of position and momentum of an electron are equal, then the uncertainity in measurement of its velocity is 7.98 x 10^{6+x} m/s. then the value of x is.

4. An ion Mn^{x+} has the magnetic moment 4.9 B.M. The value of x is.

5. A cylinder having internal volume 20 litre contains helium gas at a pressure of 2 atm and 300 K temperature. The gas is to be used to fill balloons at one atomspheric pressure and 300 K temperatrue. Number of balloons of capacity 5 litre can be filled is.

6. The maximum number of electrons that can have principal quantum number n = 3 and spin quantum number s = $-1/2$ is.

Space for rough work

Mathematics

PART – III

SECTION – A
Single Correct Choice Type

*This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. If $(1 + x + x^2)^{25} = a_0 + a_1x + a_2 x^2 + \ldots + a_{50} x^{50}$ then $a_0 + a_2 + a_4 + \ldots + a_{50}$ is:
   (A) even (B) odd and of the form 3n
   (C) odd and of the form (3n – 1) (D) odd and of the form (3n + 1).

2. A circle is touching both the coordinate axes and its centre lies o the curve $y^2 = 4x$, then area of this circle is
   (A) 16π (B) 4π
   (C) 32π (D) none.

3. The number of ordered triplet of positive integers which satisfy the inequality $20 \le x + y + z \le 50$ is
   (A) ${}^{50}C_3 - {}^{19}C_3$ (B) ${}^{50}C_2 - {}^{19}C_2$
   (C) ${}^{51}C_3 - {}^{20}C_3$ (D) none.

4. If z_1, z_2, z_3 represent the vertices of an equilateral triangle ABC and $\sqrt{\dfrac{|z_2 - z_1|^2 + |z_3 - z_1|^2}{|z_2 + z_3 - 2z_1|^2 + |z_3 - z_2|^2}} = \sin\theta$ the θ is $(0 < \theta < \pi)$
   (A) $\dfrac{\pi}{4}$ (B) $\dfrac{2\pi}{3}$
   (C) $\dfrac{\pi}{6}$ (D) $\dfrac{5\pi}{6}$.

Space for rough work

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

5. Let $\frac{x^2}{9}+\frac{y^2}{4}=1$ be a given ellipse and a bi-focal conic 'c' has its foci same as those of the given ellipse. If the conic 'c' passes through (3, 2), then its equation may be

(A) $x^2 - y^2 = 5$ (B) $\frac{x^2}{3}+\frac{y^2}{2}=1$

(C) $\frac{x^2}{15}+\frac{y^2}{10}=1$ (D) $\frac{x^2}{18}+\frac{y^2}{8}=1$

6. If the quadratic equation $x^2 + 2ax + b(a - 1) = 0$ has real roots for all real value of a and b is an integer, then which of the following is correct?

(A) greatest value of b is 4 (B) no such value of b is possible

(C) least value of b is 0 (D) b can be any integer.

7. In a ΔAEX, T is the mid point of XE, and P is the mid point of ET. If the ΔAPE is equilateral of side length equal to unity then which of the following alternative(s) is/are correct?

(A) $AX = \sqrt{13}$ (B) $\angle EAT = 90^o$

(C) $\cos(\angle XAE) = \frac{-1}{\sqrt{13}}$ (D) $AT = \frac{1}{\sqrt{3}}$.

8. The bisector of internal angles A and ΔABC meets BC in point D such that BD = 4 and CD = 2. If altitude AE > $\sqrt{10}$ and AB and AC are integers, then AC may be

(A) 4 (B) 5

(C) 6 (D) none.

Space for rough work

Comprehension Type

This section contains one group of questions. This group has 3 multiple choice question based on a paragraph. This question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 9 to 11

Read the following write up carefully and answer the following questions:
A line 4 cuts coordinate axes at A(7, 0) and B(0, –5) respectively. A variable line PQ is drawn perpendicular to AB meeting the coordinate axis at P & Q. The line AQ & BP intersect at R. T is the point on locus of R, farthest from the origin. A parabola C_1 with vertex at T and axis parallel to x-axis is drawn with lotus rectum 2 units opening towards the right of y-axis. Answer the following questions:

9. Coordinates of T are
   (A) (7,5) (B) (7, –5)
   (C) (–7, 5) (D) (–7, –5)

10. Equation of parabola C_1 is
   (A) $y^2 + 10y – 2x + 39 = 0$ (B) $y^2 + 10y + 2x + 9 = 0$
   (C) $y^2 + 10y – 2x + 9 = 0$ (D) none.

11. A light ray travelling along the line y = –3, gets reflected from the interior of the parabola, the equation of the reflected ray is
   (A) $4x + 3y = 35$ (B) $4x – 3y = – 45$
   (C) $4x – 3y = 45$ (D) none.

Space for rough work

SECTION – B

(Matching List Type)

*This section contains **2 questions.** Each question has four statements (A, B, C and D) given in **Column I** and **five statements** (p, q, r, s and t) in **Column II.** Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.*

1. There are two Indian couples, 2 American couples and one unmarried person

Column – I		Column – II	
(A)	The total numbers of ways in which they can sit in a row such that an Indian wife and an American wife are always on either side of the unmarried person	(p)	5760
(B)	The total numbers of ways in which they can sit in a row such that the unmarried man always occupy the middle position.	(q)	24230
(C)	The total number of ways in which they can sit around a circular table such that Indian wife & American wife are on either side of unmarried person.	(r)	40320
(D)	If all the nine persons are to be interviewed one by one then the total number of ways of arranging their interviews such that no wife gives interview before her husband	(s)	22680
		(t)	28410

2. Match the following Column-I with Column-II

Column – I		Column – II	
(A)	If 2sinx, sin 2x and 2 cosx are in A.P., then value of |sinx + cosx| is	(p)	$\frac{\sqrt{5}-1}{2}$
(B)	If angles of triangle are in A.P. with common difference equal to $\frac{1}{3}$ of the greatest angle then, ratio of the greater two sides is	(q)	$\frac{\sqrt{3}}{2}$
(C)	Let the sides AB, BC, CD & DA of a cyclic quadrilateral ABCD are in G.P. with common ratio 2, then $\frac{BD}{AB}$ is	(r)	$2\sqrt{5}$
(D)	Let a, b, c be in A.P. and a^2, b^2, c^2 be in H.P. If $a \neq c$, then a : c is	(s)	$\sqrt{3}-2$
		(t)	1

Space for rough work

SECTION – C

Integer Answer Type

*This section contains **6** questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.*

1. In $\Delta ABCD$, circumradius is 3 and inradius is 1.5 units. If the value of
$a\cot^2 A + b^2 \cot^3 B + c^3 \cot^4 C$ is $m\sqrt{n}$ where m and n are prime members, them $\frac{m+n}{2}$ is

2. The value of the product
$\sin\left(\frac{5\pi}{2^{11}}\right)\cos\left(\frac{5\pi}{2^{11}}\right)\cos\left(\frac{5\pi}{2^{10}}\right)\cos\left(\frac{5\pi}{2^{9}}\right)...\cos\left(\frac{5\pi}{2^{3}}\right)\cos\left(\frac{5\pi}{2^{3}}\right)$
equals to 4^{-k}. Then value of k is

3. From a point P two tangents are drawn to $\frac{x^2}{2}-\frac{y^2}{3}=1$, such that their chord of contact subtends right angle at its centre. If locus of P is the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then $\frac{6}{13}\left(a^2+b^2\right)$ is equal to

4. A circle of radius 4 units is inscribed in an equilateral triangle ABC, then an equilateral triangle is inscribed in the circle, a circle again is inscribed in the later triangle and so on. In this way the process continues infinitely. If r, x_1, x_2, … x_n … be the radii of these circles respectively. Then the sum of radii of all these circles is

5. Consider the complex number z satisfying $|z-3|+|z|+|z+3|=12$. If the greatest integer less than or equal to $|z|$ is 'a' and the least integer greater than or equal to is 'b', then 'a + b' is

6. Let $p(x) = x^5 + x^2 + 1$ have roots x_1, x_2, x_3, x_4 and x_5
$g(x) = x^2 - 2$, then the value of
$g(x_1)\, g(x_2)\, g(x_3)\, g(x_4)\, g(x_5) - 30g\,(x_1x_2x_3x_4x_5)$ is

Space for rough work